Beamer Sample

Diffusion Equation

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Agenda

Diffusion Equation

Consider the 1D diffusion equation:

$$\frac{\partial u}{\partial t} = K \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < 1,\ t > 0,\ K > 0.$$

Example boundary conditions (Dirichlet):

$$u(0, t) = u(1, t) = 0.$$

Fourier Sine Expansion

Under homogeneous Dirichlet BC, we use

$$u(x,t) = \sum_{k=1}^{\infty} \hat{u}(k,t)\,\sin(k\pi x).$$

Coefficients are obtained by

$$\hat{u}(k,t) = 2\int_0^1 u(x,t)\,\sin(k\pi x)\,dx.$$

Mode Decay

Each Fourier mode satisfies

$$\frac{d\hat{u}(k,t)}{dt} = -(k\pi)^2 K\,\hat{u}(k,t),$$

therefore

$$\hat{u}(k,t) = \hat{u}(k,0)e^{-(k\pi)^2 Kt}.$$

- Large k (fine structures) decay faster.
- The solution becomes smoother as time evolves.

Summary

1. Diffusion smooths spatial variations.
2. Fourier decomposition gives mode-wise dynamics.
3. Beamer is suitable for math-heavy slides.

Thank you!